PAID LIFT

Feature Gap Analysis & Remaining Development Report

Audit Date: October 05, 2026
Version: 1.3 Post-Build Review
Target: Production Commercial
Stack: Expo SDK 54 · FastAPI · MongoDB

1. Executive Summary & Development Completion

PAID LIFT is a peer-to-peer (P2P) ride-sharing marketplace mobile application for Indian inter-city and highway corridors. A comprehensive code and requirements audit comparing the codebase against **PRD v1.3** and **TRD v1.3** reveals that the core marketplace loop has been developed as an MVP in simulation/mock mode. However, significant production integrations, real-time infrastructure, and roadmap features remain pending before the platform can launch commercially.

Scope Layer	Completion %	Current Status	Primary Missing Elements
Core MVP Scope (PRD FR-1 to FR-8)	**75% - 80%**	**Functional MVP**	SMS OTP gateway, KYC document image upload, driver bank payouts, auto-expired rides, real GPS broadcast.
Production & Infra Readiness (Live Gateways & Scaling)	**40% - 45%**	**Simulation Mode**	Live Razorpay/PhonePe keys, Google Maps billing key, DigiLocker eKYC, WebSocket chat, Push notifications.
Total Commercial Platform (MVP + Smart Roadmap)	**50% - 55%**	**Halfway Complete**	Corridor smart matching, Driver Plus subscription, Women-only rides, Referral system, Dynamic pricing.

2. Functional Requirements Audit (PRD FR-1 to FR-8)

Detailed gap analysis of all functional requirements specified in the Product Requirements Document:

Req ID	Feature Requirement	Priority	Status	Current State & Remaining Implementation Required
FR-1.1	Phone OTP Registration	P0	**Mocked**	OTP hardcoded to 123456. Real SMS gateway (Fast2SMS/MSG91/Twilio) integration required.
FR-1.3	Aadhaar Identity Verification	P0	**Mocked**	Captures last-4 digits with mock OTP. Government DigiLocker or UIDAI eKYC API required.
FR-2.4	Vehicle Document Photo Upload	P1	**Pending**	Only text numbers stored for RC, DL & Insurance. Camera/S3 file upload missing.
FR-2.5	Admin Vehicle Verification	P1	**Pending**	Vehicles are auto-approved (approved=True). Admin document review workflow missing.
FR-3.3	Auto-Expire Past Rides	P1	**Pending**	Past rides still appear in search. Needs automated TTL / cron filter for depart_at.
FR-3.4	Ride Cancellation & Refunds	P1	**Pending**	Cancellation endpoints lack automated wallet/gateway refund calculation & ledger entries.
FR-4.9	Smart Corridor Route Matching	P1	**Pending**	Current search matches exact city regex. Spatial corridor waypoint intersection missing.
FR-4.10	Bus-Stand Pickup Selector	P1	**Pending**	Free-text input only. Verified highway bus-stop dropdowns not yet implemented.
FR-5.9	Driver Bank/UPI Withdrawal	P1	**Pending**	Owner wallet credits 90% fare, but payout flow (RazorpayX / Cashfree) is not built.
FR-6.6	Real Driver GPS Broadcast	P1	**Pending**	Live tracking uses simulated ETA timer. Needs expo-location + WebSocket broadcast.
FR-6.7	Native Google Maps Rendering	P1	**Pending**	Uses Leaflet WebView. Native react-native-maps SDK needs dev build configuration.

FR-7.5	Push Notifications	P1	**Pending**	No FCM / Expo Push tokens or trigger pipeline for booking requests, acceptances, or chat.
FR-7.7	Masked Voice Calling	P2	**Backlog**	Direct calling only. Telephony proxy (Exotel / Twilio voice) for privacy not yet wired.
FR-8.5	Revenue & Commission Analytics	P1	**Pending**	Admin panel displays simple counts. Needs GMV, net commission & ledger breakdown.
FR-8.6	Admin User/Ride Moderation	P1	**Pending**	Admin is read-only. Ban user, block ride, and resolve SOS actions are missing.

3. Technical Requirements & Architectural Debt (TRD)

Key architectural items from TRD Sections 5, 7, and 10 that must be hardened for production:

Area	Current Architecture	Target Production Architecture	Effort
Payment Gateways	Simulator mode active when keys are rzp_test_REPLACE / REPLACE.	Drop real merchant credentials in backend/.env; verify live HMAC callbacks.	Low (1-2 days)
Google Maps SDK	Mock haversine calculation with synthetic straight-line polylines.	Add billing-enabled GOOGLE_MAPS_API_KEY for real Directions & Places New.	Low (1 day)
In-App Chat	Client polls GET /api/chat every 3.0 seconds via HTTP.	Migrate to WebSockets (FastAPI WebSocket endpoint) or SSE for battery/server efficiency.	Medium (3-4 days)
Database Transactions	Single-document atomic claims (credited: true) on standalone MongoDB.	MongoDB Replica Set with multi-document ACID transactions for financial ledger.	Medium (2-3 days)
Security & Hardening	CORS allow_origins=['*'], no rate-limiting on /api/auth/send-otp.	Tighten CORS to trusted mobile/web domains; add Redis-backed token bucket rate limits.	Medium (2 days)
PhonePe Token Cache	OAuth token requested on every transaction.	Implement TTL expiry-aware memory/redis token cache as per TRD 10.3.	Low (1 day)

4. Smart Features Roadmap (PRD Section 7 — All Pending)

Planned value-add and monetization features scheduled for Phase-2:

#	Roadmap Feature	Priority	Monetization / User Impact Description
1	**Driver Plus Subscription**	**P1**	■99/month recurring plan: 0% commission on first 20 rides + priority search placement. Core recurring revenue driver.
2	**Referral & Invite Engine**	**P1**	Two-sided ■50 wallet credit on referee's first completed ride. Critical for viral user acquisition.
3	**Women-Only Ride Option**	**P1**	Female drivers can restrict passengers to verified females; female riders can filter for women-driven cars.
4	**Real-Time Ride Alerts**	**P1**	Push alerts when a driver posts a ride matching a passenger's saved route (e.g. '3 new rides Mumbai->Pune').
5	**Dynamic Pricing Advice**	**P2**	Algorithmic fare suggestions for drivers based on toll charges, distance, vehicle class, and corridor demand.
6	**Driver Gamification & Badges**	**P2**	Badges for 'Top Driver', '50 Rides Club', and 'Safe Driver' based on verified reviews and zero SOS history.
7	**EV-Only Rides Filter**	**P2**	Green travel badge for EV owners; passengers can filter specifically for eco-friendly rides.

8	**Multi-Language Localization**	P1	Full Hindi (first), followed by Marathi, Gujarati, and Kannada for Indian highway adoption.
9	**Dark Mode**	P2	System-level dark theme support tailored for night highway driving.
10	**In-Ride Micro-Insurance**	P2	Per-ride passenger personal accident cover partnership (e.g. Acko, Digit).

5. Full-Stack QA Audit & Security Findings

Code-level edge cases and business logic bugs identified during full-stack review:

Issue Location	Observed Vulnerability	Impact	Required Fix
POST /api/bookings (backend/server.py:327)	Missing validation for passenger_id == ride.owner_id.	Drivers can book seats in their own rides to artificially manipulate stats.	Add: if payload.passenger_id == ride['owner_id']: raise HTTPException(400, 'Self-booking disallowed')
PATCH /api/bookings (backend/server.py:372)	When accepting a booking, if seats_available < seats, decrement is skipped but status is still marked 'accepted'.	Overbooking occurs; UI displays negative seats available (e.g. -1 seats).	Reject status change with 400 if remaining seats cannot satisfy the booking.
CORS Configuration (backend/server.py:555)	CORS configured with allow_origins=['*'].	Cross-origin security risk in public web environments.	Restrict allowed origins to production web domain and mobile app schemes.
Public Endpoints (/auth/send-otp, /payments)	No IP rate limiting on OTP generation or payment order creation.	Vulnerable to SMS flooding and simulated order exhaustion.	Add slowapi rate limiters (e.g. 5 requests/min per IP/phone).

6. Actionable Production Implementation Plan

Recommended 4-Sprint execution sequence to bring PAID LIFT from simulation prototype to commercial launch:

Phase / Sprint	Duration	Target Deliverables	Milestone Objective
Sprint 1: Core Integrity & Secrets	1 Week	• Fix self-booking & seat overbooking bugs • Real SMS OTP gateway (Fast2SMS/MSG91) • Configure live Razorpay & PhonePe keys • Configure Google Maps API billing key	**End-to-End Real Transactions**
Sprint 2: Essential Operations	1.5 Weeks	• Vehicle document photo upload to Cloud Storage • Admin vehicle approval/rejection panel • Owner bank/UPI payout flow (RazorpayX) • Ride cancellation & automatic refund engine	**Trust & Financial Closure**
Sprint 3: Real-Time & Communications	1.5 Weeks	• Push Notifications (Expo Push / FCM) • WebSocket protocol for in-app chat • Background GPS location tracking (driver) • Admin moderation actions (suspend/delist)	**Live Experience Ready**
Sprint 4: Growth & Commercialization	2 Weeks	• Smart corridor waypoint matching • Driver Plus subscription engine (■99/mo) • Two-sided referral reward system • Hindi language localization	**Commercial Market Launch**